

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1437

Clairemont 345 kV
June 21, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1437. The Interconnection Customer proposes a 184.3 MW solar photovoltaic facility interconnecting at the Clairemont 345 kV substation in the Kentucky area. The request was received on March 23, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 494 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$61,145,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Yuba City Sta A - Frontier 138 KV
Contingency: McClellan Ar - Spatsum Lornex 138 KV
Loading: 125.9% (Rating: 405 MVA)
DFAX: 77.1%
Pre-project loading: 90.8%
2. Facility: Kettle - Country Club 2 138 KV
Contingency: Spatsum Lornex - Latah Jct 138 KV line
Loading: 125.4% (Rating: 154 MVA)
DFAX: 32.5%
Pre-project loading: 86.5%
3. Facility: Yuba City Sta A - Westin-Elgin 138 KV line
Contingency: McClellan Ar - Spatsum Lornex 138 KV line
Loading: 124.9% (Rating: 418 MVA)
DFAX: 77.1%
Pre-project loading: 90.9%
4. Facility: Westin-Elgin - McClellan Ar Substation 138 KV
Contingency: Yuba City Sta A - Westin-Elgin 138 KV
Loading: 121.8% (Rating: 359 MVA)
DFAX: 59.7%
Pre-project loading: 91.1%
5. Facility: McClellan Ar - Spatsum Lornex 138 KV
Contingency: Yuba City Sta A - Westin-Elgin 138 KV
Loading: 118.4% (Rating: 464 MVA)
DFAX: 81.4%
Pre-project loading: 86.1%

6. Facility: Temple Sta 243 - Westin-Elgin 138 KV
 Contingency: McClellan Ar - Westin-Elgin 138 KV
 Loading: 118.4% (Rating: 300 MVA)
 DFAX: 44.4%
 Pre-project loading: 91.1%
7. Facility: Harvest Power Orlando - Frontier 138 KV
 Contingency: Latah Jct - Spatsum Lornex 138 KV
 Loading: 117.3% (Rating: 304 MVA)
 DFAX: 50.6%
 Pre-project loading: 86.6%
8. Facility: Country Club 2 - Spatsum Lornex 138 KV
 Contingency: Spatsum Lornex - Latah Junction 138 KV line
 Loading: 111.5% (Rating: 179 MVA)
 DFAX: 32.5%
 Pre-project loading: 78.0%
9. Facility: McClellan Ar Substation - Temple Station 243 138 KV
 Contingency: Westin-Elgin - McClellan Ar 138 KV
 Loading: 109.8% (Rating: 295 MVA)
 DFAX: 44.4%
 Pre-project loading: 82.0%
10. Facility: Frontier - Delano 138 KV
 Contingency: Spatsum Lornex - McClellan Ar 138 KV
 Loading: 100.1% (Rating: 176 MVA)
 DFAX: 29.7%
 Pre-project loading: 69.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yuba City Sta A - Frontier 138 KV	Rebuild line with higher-capacity conductor	\$13,076,000	62% (\$8,039,000)
Kettle - Country Club 2 138 KV	Rebuild line with higher-capacity conductor	\$11,015,000	77% (\$8,504,000)
Yuba City Sta A - Westin-Elgin 138 KV line	Rebuild line with higher-capacity conductor	\$13,796,000	62% (\$8,482,000)
McClellan Ar - Westin-Elgin Substation 138 KV Line	Rebuild line with higher-capacity conductor	\$12,171,000	61% (\$7,420,000)
McClellan Ar - Spatsum Lornex 138 KV	Rebuild line with higher-capacity conductor	\$11,358,000	61% (\$6,925,000)
Temple Sta 243 - Westin-Elgin 138 KV	Rebuild line with higher-capacity conductor	\$3,009,000	59% (\$1,780,000)
Harvest Power Orlando - Frontier 138 KV	Rebuild line with higher-capacity conductor	\$20,104,000	48% (\$9,718,000)
Spatsum Lornex - Country Club 2 138 KV	Rebuild line with higher-capacity conductor	\$9,155,000	77% (\$7,069,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Temple Station 243 - McClellan Ar 138 KV	Reconductor line	\$2,716,000	59% (\$1,607,000)
Frontier - Delano 138 KV	Reconductor line	\$2,054,000	78% (\$1,601,000)
	Total allocated to project		\$61,145,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.